Shadman Salif Swanan

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Shad1573 | Shadman Salif Swanan

EDUCATION

BRAC University 2022 – 2026
Bachelor of Science in Computer Science and Engineering (CSE) Dhaka, Bangladesh
- **CGPA**: 3.83 out of 4.00 (Merit based Scholarship)
- **Programming Coursework**: Fundamentals of Programming, Object-Oriented Programming, Data Structures & Algorithms
- **Computational Coursework**: Natural Language Processing, Artificial Intelligence, Computer Network, Linear Algebra, Neural Networks, Digital Logic Design, VLSI Design, Discrete Mathematics

Notre Dame College 2018 – 2021
Higher Secondary Certificate (HSC), Science Dhaka, Bangladesh
- **GPA**: 5.00 out of 5.00

Ideal School and College 2008 – 2018
Secondary School Certificate (SSC), Science Dhaka, Bangladesh
- **GPA**: 5.00 out of 5.00 (General Scholarship/ Dhaka Board)

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, PHP
Data Analysis: MySQL, Power BI, Microsoft Excel (Advanced), Process Modeling, EDA
AI/ML Frameworks: PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Tensorflow, Hugging Face
Web & Tools: React, Vite, Node.js, Express.js, MongoDB, Git & GitHub, Tailwind CSS, Figma
SQA & Testing: Postman (API Testing), Test Case Design
Language Proficiency: English, Bengali

PROJECTS

- **Predictive Telecom Churn Optimization Framework** Engineered a machine learning pipeline processing 100K customer logs with an advanced Gradient Boosting model, achieving a 0.6900 ROC-AUC score and 64.00% churn recall to project an estimated 2.22M USD in annualized retention savings.
- **Real-Time Fraud Detection System** Developed an end-to-end ML pipeline using LightGBM and cost-sensitive threshold tuning to detect credit card anomalies within highly imbalanced transaction streams.
- **Leaf Disease Classification (TinyResNet)** Designed a lightweight convolutional neural network with distortion-aware augmentations, reaching 96.59% classification accuracy under real-world distortions.
- **Token-Level POS Tagging using RNN Models** Implemented sequence labeling frameworks comparing RNN, LSTM, GRU, and BiLSTM architectures with 300D word embeddings to achieve 97% accuracy via BiLSTM.
- **Disease Prediction using Machine Learning** Developed classification models for diabetes prediction using KNN, Decision Tree, and Naive Bayes on clinical data after MinMaxScaler feature mapping.
- **Shikkha.com** Built a full-stack EdTech platform featuring JWT access limits, secure Stripe integration, Automated Testing consoles, and AI-assisted Mistral API educational enhancements.
- **Federal Flight Agency Mission-Critical Aerospace Network Design** Designed a secure and scalable aerospace network architecture using VLSM and floating static routing profiles for automatic mission failover.
- **3D Car Racing Game** Built a 3D racing game using OpenGL with enemy AI, shooting mechanics, weather effects, and camera modes.
- **Automated Sanitizer and Mask Dispensing Robot** Embedded systems project integrating contactless automated sanitizer pumping mechanics and proximity sensors using Arduino Uno.

RESEARCH

Undergraduate Thesis Research 2025 – 2026
BRAC University
- Conducting research on **semantic drift in the Bengali language** using diachronic word embeddings and transformer-based representations.
- Developing a column detection model for scanned Bengali newspapers to improve document layout understanding.

- Building an OCR pipeline to construct a clean and high-quality diachronic text corpus for downstream NLP tasks.
- Performed **Exploratory Data Analysis (EDA)** to quantify OCR error distributions and validated post-processing techniques that reduced linguistic noise across a 120-year diachronic corpus.
- Fine-tuned a **Historical BanglaBERT** model on a curated diachronic corpus and analyzed semantic drift using contextual embeddings.

EXPERIENCE

University of Tokyo (Matsuo–Iwasawa Laboratory) Apr. 2026 – Present

GCI World Program (Global Consumer Intelligence)

- Participating in an international AI education program focused on data science and machine learning.
- Engaging with global participants to develop practical and theoretical understanding of AI applications.

Grameenphone Academy Mar. 2026

UX Design Decision-Making with AI & Design System Thinking with AI

- Completed certifications focusing on AI-assisted user-centered design and structured design system thinking.
- Explored how AI enhances product design, decision-making, and scalable system development.

CO-CURRICULAR ACTIVITIES

Notre Dame Nature Study Club

Vice President, Department of Photography

- Led photography initiatives and coordinated visual documentation for club activities and events.
- Managed team members and contributed to creative direction and event coverage.

Notre Dame College Model United Nations

Conference Management Officer

- Managed responsibilities in the Conference Management Department during a large-scale academic event.
- Demonstrated strong organizational skills, adaptability, and ability to work efficiently under pressure.

BRAC University Residential Semester

Stage Drama Course Participant

- Completed an intensive stage drama course emphasizing teamwork, communication, and performance skills.

BRAC University Sports Activities

Football Achievements

- Champion & Top Scorer, Intra-Spring 2022 Football Tournament, BRAC University.
- Runners-Up, Crusade 2026 (Semester Tournament) & RS Cup.
- Runners-Up, Residential Semester (RS) Football Tournament.

REFERENCES

Dr. Chowdhury Mofizur Rahman
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